MERITS: Medication Recommendation for Chronic Disease with Irregular Time-Series

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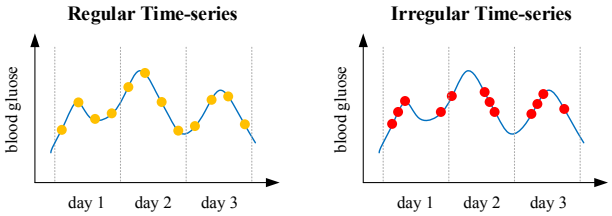
***Abstract*—Medication recommendation for chronic diseases based on the complex historical electronic medical records (EMR) is an important and challenging research problem in medical informatics because the medical records are often irregularly sampled and contain many missing data. However, most existing approaches fail to explore the irregular time-series dependencies and ignore the consecutive correlation in dynamic prescription history. To fill this gap, we propose the MEdication Recommendation network on Irregular Time-Series (MERITS), which captures the irregular time-series dependencies with the neural ordinary differential equations (Neural ODE). Meanwhile, it leverages a drug-drug interaction knowledge graph and two learned medication relation graphs to explore the co-occurrence and sequential correlations of the medications. We further propose an attention-based encoder-decoder framework to combine the historical information of patients and medications from EMR. Besides, we collect and annotate a diabetes inpatient medication dataset and demonstrate the effectiveness of MERITS by comparing it with several state-of-the-art methods of medication recommendations.**

***Index Terms*—medication recommendation, irregular time-series, sequence representation, continuous adaptation**

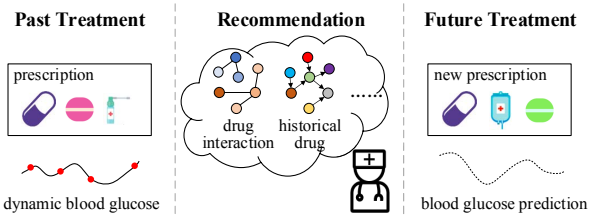
I. Introduction

With the help of powerful deep neural networks, in recent years, clinicians and data scientists try to solve challenging medical problems based on rich medical data, such as Electronic Medical Records (EMR) [1]. Among them, the medication recommendation for chronic diseases, such as diabetes, is a complex and challenging task for both clinicians and data scientists because the management of such chronic disease requires frequent monitoring of patient dynamic status and consideration of patient's underlying health status like demographic, co-morbidity and disease history. Therefore, doctors usually make prescriptions with multiple medicines with different effects and make adjustments timely based on the change of patient status. Such medication adjustment requires a lot of experience, where doctors need to consider the effectiveness and interactions of drug mixing. An automated drug recommendation model for chronic disease trained on massive EMR data can guide junior doctors for treatment planning and enable timely medication adjustment for numerous self-monitored outpatients with chronic diseases. However, there are several challenges for such recommendation models:

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(a) Time-series measurement of diabetic patients' blood glucose.



(b) An example of the medical recommendation procedure.

Fig. 1. Illustration of the irregular time-series problem in EMR and example of the medical recommendation procedure.

(1) How to represent the irregular-sampled time-series health data of patients and combine this dynamic information with underlying static health conditions? (2) How to provide a longitude time-series medication recommendation and timely adjust medication based on patients' reaction to the medicine? (3) How to integrate the knowledge on drug interaction and retrieve doctors' experience on medicine combination and sequence?

To address these challenges, we propose a novel model - *Medication Recommendation Network on Irregular Time-Series* (MERITS), which captures the irregular time-series dependencies with the neural ordinary differential equations (Neural ODE) [2] and leverage the knowledge of drug-drug interaction and medication relation. Specifically, we focus on a common chronic disease, diabetes. As shown in Figure 1(a), blood glucose, the dynamic time-series feature of diabetic patients, can vary significantly within a day, and the values are irregular-sampled over time. While existing methods deal with irregularity using RNN-based models, which intrinsically

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work for discrete sequence, our proposed MERITS models the irregular time-series dependencies with Neural ODE, which can better model the continuous inner process. MERITS further leverages the encoder-decoder architecture to predict the following medication sequence and integrates the static and dynamic features by self-attention. Meanwhile, MERITS use three graphs to embed and leverage the knowledge of drugs and doctor experience. As illustrated in Figure 1(b), the medicines are prescribed in groups and managed over time based on the dynamic glucose level, static underlying conditions, drug interactions and doctors' experience.

Furthermore, while many medication recommendation models [3]–[7] define their task as a per-visit medication recommendation for outpatient with a short sequence (sequence length of 2 to 3 on average) or for inpatients over a short duration (24 or 48 hours), we collect and annotate a dataset in a more realistic scenario for inpatients with chronic diseases, where medication is prescribed daily during the hospitalization (around 14 days). With such data, we define our medication recommendation task with longer time steps and more frequent prescriptions. Empirically, we extensively demonstrate the effectiveness of MERITS by comparing it with several state-of-the-art methods. Our case study further investigates the interpretability of our model by explaining why some medicine is recommended and how the recommended medicine changes over time. The contributions of this paper can be summarized as follows:

- Inspired by Transformer [8], we propose a sequential medication recommendation framework, MERITS, based on the Encoder-Decoder architecture.
- We develop Neural-ODE and self-attention mechanism to integrate irregular dynamic time-series data and static features.
- We design graph-augmented attention to incorporate medical knowledge on drug-drug interaction and doctors' experience on medicine sequence and co-occurrence.
- We collect a irregular time-series diabetic inpatient medication recommendation dataset with daily dynamic prescriptions and longer sequences.

II. Problem Formulation

Electronic Medical Records (EMR) document the disease progression of each patient, and the records are long, complex historical observations. The EMR data consist of three parts: (a) the demographics, such as height, weight, age, etc., which are referred to as "static features"; (b) the bio-markers, like blood pressure, blood oxygen, glucose, etc., which reflect the patient's status over time, so we call them "dynamic features"; (c) the medications prescribed at each time stamp. Therefore, the medical record is defined as:

Definition 1: **Patient medical record**. A patient's observation at timestamp t is defined as: $\boldsymbol{x}^t = [\boldsymbol{c}_s, \boldsymbol{c}_d^t, \boldsymbol{c}_m^t]$, where $t \in \{1, 2, \ldots, T\}$, and T is the total observation length. $\boldsymbol{c}_s \in \mathbb{R}^S$ represents S static features of the patient. $\boldsymbol{c}_d^t = \left[\boldsymbol{c}_{d_1}^t, \boldsymbol{c}_{d_2}^t, \ldots, \boldsymbol{c}_D^t\right]$ represents D dynamic features. $\boldsymbol{c}_{d_i}^t \in \mathbb{R}^{\mathcal{T}^t}$ describes the changing of i-th dynamic feature within the period of observation at timestamp t, where $\mathcal{T}^t$ is an uncertain number stands for the number of sampling points of the dynamic feature within the period, which generates the irregular time-series. $\boldsymbol{c}_m^t \in \{0, 1\}^{|\mathcal{M}|}$ is the medication record coding for $|\mathcal{M}|$ medications.

The medical recommendation problem is formulated as:

Problem 1: **Medication recommendation problem**. Given a patient's medical record sequence $\boldsymbol{\mathcal{P}} = \left[\boldsymbol{x}^1, \boldsymbol{x}^2, \ldots, \boldsymbol{x}^t\right]$ of length t, the problem is to determine the next medication combination for this patient at timestamp $t+1$.

III. Method

A. Overview of The MERITS

Our MERITS holds an encoder-decoder architecture, which is shown in Figure 2. The **Encoder** consists of a medical embedding module, a dynamic encoding module and a patient aggregation module. It captures the medication strategies and the underlying status of patients from an irregular sampled time-series, generates the representation of the patient at current timestamp. The **Decoder** consists of a graph attention module and a medication generator. It captures relationship between drugs in the medication history with graphs and generates medication recommendation at timestamp $t+1$ based on the patient representation.

B. Encoder: Representing Patient's State Under Irregular Time-Series Observations

1) Medication Embedding Module: The medication history is observed as a sequence of multi-hot vectors. Given a series of patient's medication observations $\left[\boldsymbol{c}_m^1, \boldsymbol{c}_m^2, \ldots, \boldsymbol{c}_m^t\right]$, we first capture the sequential information using the Recurrent Neural Network (RNN) [9], $\boldsymbol{h}_m^t = \sigma\left(\boldsymbol{U}_m \widehat{\boldsymbol{c}_m^t} + \boldsymbol{W}_m \boldsymbol{h}_m^{t-1} + b_m\right)$. $\boldsymbol{h}_m^t$ denotes the hidden state of the medication coding at timestamp t. Thus, the patient's historical medication coding are represented as $\boldsymbol{h}_m = \left[\boldsymbol{h}_m^1, \boldsymbol{h}_m^2, \ldots, \boldsymbol{h}_m^t\right]$.

Then the hidden states are fed into a self-attention module with $\boldsymbol{Q}_m = \boldsymbol{W}_m^Q \boldsymbol{h}_m$, $\boldsymbol{K}_m = \boldsymbol{W}_m^K \boldsymbol{h}_m$, $\boldsymbol{V}_m = \boldsymbol{W}_m^V \boldsymbol{h}_m$:

$$\mathcal{A}(\boldsymbol{Q}_m, \boldsymbol{K}_m, \boldsymbol{V}_m) = \mathrm{Softmax}\left(\frac{\boldsymbol{Q}_m \boldsymbol{K}_m^T}{\sqrt{d_{k_m}}}\right) \boldsymbol{V}_m, \qquad (1)$$

where $\boldsymbol{W}_m^Q, \boldsymbol{W}_m^K, \boldsymbol{W}_m^V$ is a trainable weight matrix that transfer the inputs $\boldsymbol{h}_m$ into *query*, *key* and *value* matrix. The output is projected with a linear layer and produce the patient's medication embedding $\boldsymbol{E}_m$.

2) Dynamic Encoding Module: It is crucial to capture and represent the irregular time-series dynamic features to depict the patient's status at time t. Based on this, we can determine what is the best medication strategy for this patient at time $t+1$. The neural ordinary differential equation (Neural ODE) is an efficient method to model the time-series data. We design it with a pre-training and fune-tuning process on the dynamic features to represent the irregular time-series.

We first utilize a Neural ODE to model the irregular time-series data in a certain timestamp. It can be viewed as there is

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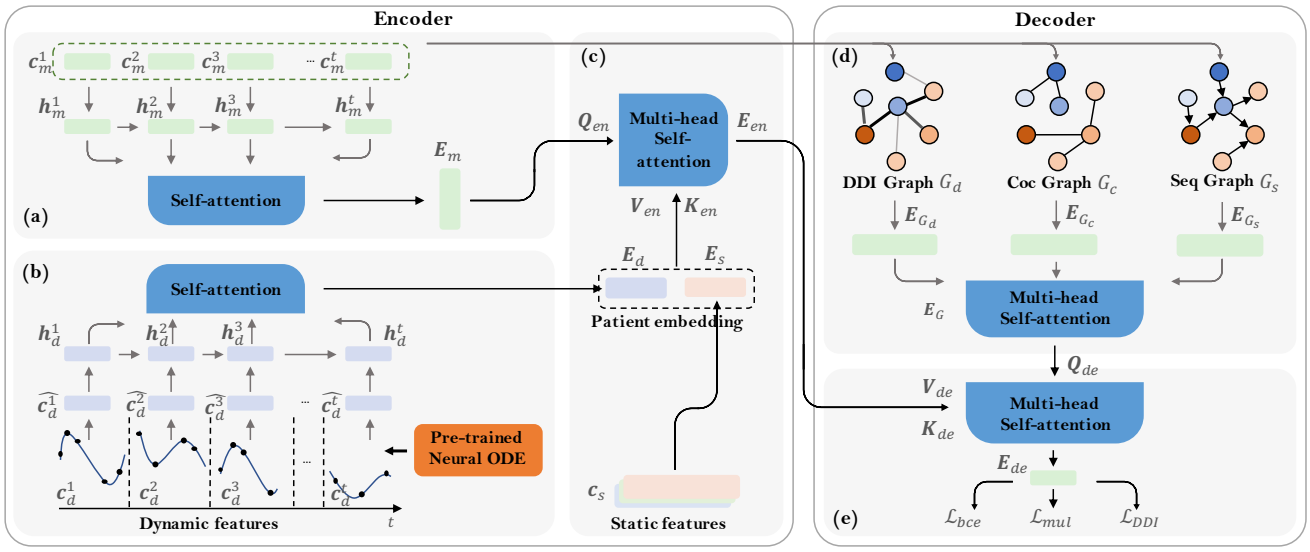


Fig. 2. The overview architecture of MERITS. **Encoder:** (a) medical embedding module; (b) dynamic encoding module; (c) patient aggregation module; **Decoder:** (d) graph attention module; (e) medication generator.

a process following an unknown ODE and some observations $c_{d_i}^t$ at τ_i along its trajectory. Denote the hidden state of the Neural ODE at τ_i as $z(\tau_i)$, we have $\frac{dz}{d\tau} = f(z(\tau), \tau)$.

The goal is to use a Neural ODE $\widehat{f}_\theta(z(\tau), \tau)$ with parameters denoted as θ to approximate the dynamic function $f(z(\tau), \tau)$. Thus we have $z(\tau) = z(\tau_0) + \int_{\tau_0}^{\tau} \widehat{f}_\theta\left(z(s), s\right) \mathrm{d}s, z(\tau_0) = \ell_\theta(c_{d_1}^t)$, where $\ell_\theta(\cdot)$ is the linear function with the parameters θ that maps the original inputs to a initial hidden state of the ODE.

Then the hidden states of the Neural ODE can be written into a continuous form $\{z(\tau)\}_{\tau=\tau_0}^{\tau_t}$, which includes the information of the whole observed trajectory of the timestamp t. We sample several empirical regular time points $\{\tilde{\tau}_1, \tilde{\tau}_2, \tilde{\tau}_3, \ldots\}$ in the continuous hidden state to represent the hidden state of the dynamic features of patient at timestamp t in a vector form, which is denoted as $\widehat{\boldsymbol{c}}_d^t$. Thus MERITS converts the irregular time-series observations of each timestamp into a fixed-size representation. In this way, the dynamic feature embedding is a fixed-representation of all historical observations in the irregular time series, and the dynamic embeddings of each timestamps is denoted as a sequence $\left[\widehat{\boldsymbol{c}}_d^1, \widehat{\boldsymbol{c}}_d^2, \ldots, \widehat{\boldsymbol{c}}_d^t\right]$. Similar to the medication encoding module, the dynamic features embeddings are fed into the RNN and self-attention layer to capture the underlying sequential features.

3) Patient Aggregation Module: This module models the patient's state by aggregating the sequential medications, the dynamic features, and the static features. For static features $\boldsymbol{c}_s$, we compute the embedding using a simple linear map: $\boldsymbol{E}_s = \mathrm{linear}(\boldsymbol{c}_s)$, where $\boldsymbol{E}_s \in \mathbb{R}^{1\times d_s}$. Then $\boldsymbol{E}_s$ is copied t times, stacked and concatenated with the dynamic embedding. Then $\boldsymbol{E}_m$, $\boldsymbol{E}_s$ and $\boldsymbol{E}_d$ are fed into a multi-head self-attention block to capture the dependence between medication and patient state. Note that the *query* is defined as $\boldsymbol{Q}_{en} = \boldsymbol{W}_{en}^Q \boldsymbol{E}_m$, the *key* and *value* matrices are defined as $\boldsymbol{K}_{en} = \boldsymbol{W}_{en}^K [\boldsymbol{E}_s, \boldsymbol{E}_d]$, $\boldsymbol{V}_{en} = \boldsymbol{W}_{en}^V [\boldsymbol{E}_s, \boldsymbol{E}_d]$, respectively, $\boldsymbol{W}_{en}^Q \in \mathbb{R}^{n\times d_m \times d_{en}}$, and $\boldsymbol{W}_{en}^K, \boldsymbol{W}_{en}^V \in \mathbb{R}^{n\times(d_s+d_d)\times d_{en}}$. The output $\boldsymbol{E}_{en}$ of the *Encoder* is then computed as:

$$\begin{aligned} \mathcal{MA}(\boldsymbol{Q}_{en}, \boldsymbol{K}_{en}, \boldsymbol{V}_{en}) &= [head_1, \ldots, head_n]\boldsymbol{W}_{en}^O, \\ head_i &= \mathcal{A}\left(\boldsymbol{Q}_{en}^i, \boldsymbol{K}_{en}^i, \boldsymbol{V}_{en}^i\right),\ i = 1, \ldots, n \end{aligned} \tag{2}$$

where $\boldsymbol{W}_{en}^O \in \mathbb{R}^{(n*t)\times d_{en}}$ is a trainable weight matrix maps the multi-head output into a fixed size matrix $\boldsymbol{E}_{en} \in \mathbb{R}^{t\times d_{en}}$.

C. Decoder: Generating Medication Recommendation with Graph Augmentation

1) Graph Attention Module: We first build three graphs: DDI graph, Coc graph and Seq graph, to represent drug conflicts, drug co-occurrences, and sequential relationships of drugs, respectively. The nodes in the graphs are candidate drugs that can be selected for medication, and the edges in the three graphs stand for three different relationships among drugs. Based on these graphs, we integrate the historical medication into the features of the corresponding nodes. The attribute matrix of an arbitrary graph at time $t-1$ is denoted as $\boldsymbol{G}_*^{t-1} \in \mathbb{R}^{|\mathcal{M}|\times k}$. Here k stands for the dimension of node attributes. We only use one node attribute that is the usage times of a drug, so we have $k = 1$. The attribute matrix is accumulated by the true medication vector at each timestamp: $\boldsymbol{G}_*^t = \boldsymbol{G}_*^0 + \sum_{i=1}^t \boldsymbol{c}_m^i$, where $\boldsymbol{G}_*^0$ is the initial attribute matrix of the graphs.

To get sufficient expression ability to transform input features into representative features, with the attributed graphs $\mathcal{G}_d = (\boldsymbol{G}_*^t, \boldsymbol{A}_d)$, $\mathcal{G}_c = (\boldsymbol{G}_*^t, \boldsymbol{A}_c)$ and $\mathcal{G}_s = (\boldsymbol{G}_*^t, \boldsymbol{A}_s)$ at time t, we perform Graph Attention operation (GAT) [10] on the attribute matrix: $\boldsymbol{G}_*^t = \{\vec{v_1}, \vec{v_2}, \ldots, \vec{v_{|\mathcal{M}|}}\}$, where $\vec{v_i} \in \mathbb{R}^k$, and we compute the attention coefficient between drug i and drug j as

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$$\alpha_{ij*} = \frac{\exp\left(\sigma_L\left(\vec{a}^\top\left[\mathbf{W}\vec{v}_i \| \mathbf{W}\vec{v}_j\right]\right)\right)}{\sum_{k\in\mathcal{N}_{i*}}\exp\left(\sigma_L\left(\vec{a}^\top\left[\mathbf{W}\vec{v}_i \| \mathbf{W}\vec{v}_k\right]\right)\right)}, \tag{3}$$

where $\mathcal{N}_{i*}$ refers to the neighbor of drug i in the three graphs, $\mathbf{W} \in \mathbb{R}^{d\times k}$ is a linear transformation matrix, and $\vec{a} \in \mathbb{R}^{1\times 2d}$ is a vector that map the concatenate of vectors into scalars, σ_L is the LeakyReLU [11] activation function. Then the multi-head attention is performed as $\vec{v}'_{i*} = \|_{k=1}^{K}\sigma\left(\sum_{j\in\mathcal{N}_{i*}}\alpha_{ij*}^k\mathbf{W}^k\vec{v}_j\right)$, where K denotes the number of attention heads, $\|$ denotes the concatenate operation over all results.

Thus we get the output of the graph attention module, which is denoted as $\boldsymbol{E}_{G_*} = \{\vec{v'_{1*}}, \vec{v'_{2*}}, \ldots, \vec{v'_{|\mathcal{M}|_*}}\}$, $\vec{v'_i} \in \mathbb{R}^K$. To capture the attendance of drugs between the three graph, the three graph embeddings $\boldsymbol{E}_{G_d}, \boldsymbol{E}_{G_c}, \boldsymbol{E}_{G_s} \in \mathbb{R}^{|\mathcal{M}|\times K}$ are then fed into multi-head self-attention layer and computed as Equation 2. Finally, we get the output $\boldsymbol{E}_G$, the overall embedding of the three graphs which integrate the historical medication of the graph attention module.

2) Medication Generator: The medication generator accepts the output embedding $\boldsymbol{E}_{en}$ of the *Encoder* and the output embedding $\boldsymbol{E}_G$ of the graph attention module in the *Decoder*, which represents the patient-medication attentive representation and medication-medication attentive representation, respectively. We utilize a multi-head self-attention to capture and fusion the two embeddings and produce the output embedding $\boldsymbol{E}_{de}$, which is maps to the final output $\widehat{\boldsymbol{c}_m^{t+1}}$ of the *Decoder* by a fully connected layer. The detailed implementations are described below.

The two input embeddings are first projected into three important matrices, the *query*, *key*, and *value* matrix by a linear map. Specifically, the medication-medication attentive representation $\boldsymbol{E}_G$ is treated as the *query*: $\boldsymbol{Q}_{de} = \boldsymbol{W}_{de}^Q\boldsymbol{E}_G$, where $\boldsymbol{Q}_{de} \in \mathbb{R}^{|\mathcal{M}|\times d_{de}}$; the patient-medication attentive representation $\boldsymbol{E}_{en}$ is treated as *key* and *value*: $\boldsymbol{K}_{de} = \boldsymbol{W}_{de}^K\boldsymbol{E}_{en}$, $\boldsymbol{V}_{de} = \boldsymbol{W}_{de}^V\boldsymbol{E}_{en}$, where $\boldsymbol{K}_{de} \in \mathbb{R}^{t\times d_{de}}$, and $\boldsymbol{V}_{de} \in \mathbb{R}^{t\times d_{de}}$. In this way, the current elements (drug representations) in *query* can attend to every timestamp representations of the patient-medication state. The output $\boldsymbol{E}_{de}$ of the *Decoder* is then computed as:

$$\begin{aligned}\mathcal{MA}(\boldsymbol{Q}_{de}, \boldsymbol{K}_{de}, \boldsymbol{V}_{de}) &= [head_1, \ldots, head_n]\boldsymbol{W}_{de}^O \\ head_i &= \mathcal{A}\left(\boldsymbol{Q}_{de}^i, \boldsymbol{K}_{de}^i, \boldsymbol{V}_{de}^i\right),\ i = 1, \ldots, n,\end{aligned} \tag{4}$$

where $\boldsymbol{W}_{de}^O \in \mathbb{R}^{(n*t)\times d_{de}}$ is a trainable weight matrix that maps the multi-head output into a fixed size matrix $\boldsymbol{E}_{de} \in \mathbb{R}^{t\times d_{de}}$. Finally, the medication probability vector is generated as the output of the *Decoder*: $\widehat{\boldsymbol{y}} = \widehat{\boldsymbol{c}_m^{t+1}} = \mathrm{linear}(\boldsymbol{E}_{de})$.

D. Training of the MERITS

1) Pre-training of The Dynamic Encoding Module: Before training the whole MERITS, we first pre-train the Neural ODE on the training set, which leads to the pre-trained Neural ODE can capture the latent variation information underlying all the patients in the training set. We consider the pre-training task as an ODE initial value problem.

Problem 2: **ODE initial value problem.** Consider the task that there are only 2 observations, at the beginning and the end of the trajectory, which is denoted as $(z_0 = z(\tau_0), \tau_0)$, $(z_1 = z(\tau_1), \tau_1)$, respectively. We starts the evolution of the system from (z_0, τ_0) for time $\tau_1 - \tau_0$ with the dynamic function f_θ that parameterized by θ using any ODE initial value solver. Then we can get a new hidden state $\widehat{z_1}$ at time τ_1. The goal of optimize the ODE is to minimize the difference between the estimation $\widehat{z_1}$ and observation z_1 by tuning the parameters θ.

In the pre-training, we follow the ODE initial value problem, starts with the initial value z_0 and compute the evolution to each observation $\boldsymbol{c}_d^\tau$ at time τ. Then compute the difference and make back-propagation using gradient descent following the loss function:

$$\begin{aligned}\mathcal{L}_{ode} &= \mathcal{L}\left(\int_{\tau_0}^{\tau_1} f(z(\tau), \tau, \theta)d\tau\right) \\ &= \mathcal{L}\left(\ \mathrm{ODESolve}\left(z(\tau_0), f, \tau_0, \tau_1, \theta\right)\ \right)\end{aligned}, \tag{5}$$

The details of solving the gradient have followed the algorithm in [2]. We omit the description here. Note that ODE is dependent on the initial value z_0. In this paper, we set z_0 as a function of the observation values within the current timestamp.

2) Construction of The Graphs: Here we show how we construct the three graphs, DDI graph, Coc graph and Seq graph.

DDI graph indicates the conflict relationships between pairs of drugs. With a public available website[1], which reports massive drug conflict events. We search and record the conflict between the drugs in the drug set $\mathcal{M}$ on the website. We build the adjacency matrix $\boldsymbol{A}_d$ of the DDI graph based on the frequencies reported, where higher weights on the edges reflect more conflict cases. Then the edge weights in the adjacency matrix are normalized as the final graph structure.

The construction of the Coc graph and Seq graph are treated as a pre-task of MERITS, which indicates the degree of the co-occurrence relationship and the sequential medication relationship in the history over all medications, respectively. We build these two graphs by the statistics on the whole training set. $\boldsymbol{A}_c^{ij}, \boldsymbol{A}_c^{ji}$ is weighted by the co-occurrence times of drug i and drug j. $\boldsymbol{A}_s^{ij}$ is also weighted by the sequential medication record times where drug i is used before drug j. Note that the adjacency matrix $\boldsymbol{A}_c$ of Coc graph is symmetric. However, the adjacency matrix $\boldsymbol{A}_s$ of Seq graph is non-symmetric.

3) Combined Loss Function: Follow [4], we use three loss functions to guide the training. For multi-label prediction, the two commonly used loss functions are binary cross entropy loss $\mathcal{L}_{bce}$ and the multi-label margin loss $\mathcal{L}_{mul}$, where $\mathcal{L}_{mul}$ can make the predicted probability of ground truth labels has at least 1 margin larger than other, which will enhance the difference between true labels and others. $\mathcal{L}_{bce} = -\sum_i \boldsymbol{y}_i \log\sigma\left(\hat{\boldsymbol{y}}_i\right) + (1-\boldsymbol{y}_i)\log\left(1-\sigma\left(\hat{\boldsymbol{y}}_i\right)\right)$, $\mathcal{L}_{mul} =$

[1]https://fis.fda.gov/sense/app/d10be6bb-494e-4cd2-82e4-0135608ddc13/sheet/33a0f68e-845c-48e2-bc81-8141c6aaf772/state/analysis

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$\sum_{ij} \frac{\max(0,1-(\hat{\boldsymbol{y}}[\boldsymbol{y}[j]]-\hat{\boldsymbol{y}}[i]))}{|\mathcal{M}|}$ $(0 \leqslant \boldsymbol{y}[j] \leqslant |\mathcal{M}|, i \neq \boldsymbol{y}[j])$. The two multi-label prediction losses are combined as $\mathcal{L}_m$ with a trade-off factor λ,

$$\mathcal{L}_m = \lambda \mathcal{L}_{bce} + (1-\lambda)\mathcal{L}_{mul}. \tag{6}$$

For drug safety, the DDI loss is also used which controls drug conflicts in the recommendation: $\mathcal{L}_{DDI} = \sum_{i,j} \left(\boldsymbol{A}_d \odot \left(\hat{\boldsymbol{y}}_t^\top \hat{\boldsymbol{y}}_t \right) \right)[i,j]$, where $\odot$ is the element-wise product. With the simulated annealing which proposed by [12], the overall loss function is a combination of the multi-label prediction loss and DDI loss:

$$\mathcal{L} = \begin{cases} \mathcal{L}_m & \text{if } s' \leq s \\ \mathcal{L}_{DDI}, \text{ with prob. } p = \exp\left(-\frac{s'-s}{Temp}\right) & \text{if } s' > s \\ \mathcal{L}_m, \text{ with prob. } p = 1 - \exp\left(-\frac{s'-s}{Temp}\right) & \text{if } s' > s \end{cases}, \tag{7}$$

where s' is the current DDI rate, s is our tolerance DDI rate, p is the probability that the corresponding loss will be used in the current training step, $Temp$ refers to the temperature, which with a decay rate ϵ : $Temp \leftarrow \epsilon Temp$. With this strategy, $\mathcal{L}_{DDI}$ is more likely to be used when the model outputs a high DDI rate that is larger than the tolerance s, which means the drug recommended is not safe enough. Furthermore, the model becomes more stable with the training process, $Temp$ will decay to a small value. Thus the multi-label prediction loss $\mathcal{L}_m$ is more likely to be used to constrain the model to pay more attention to the multi-label prediction task.

IV. Experiments

A. Data Description

We collected a real-world diabetic inpatient medication dataset containing 1,300 diabetes inpatients from late 2014 to early 2020. Around 43% of the patients are female. The average age is 62.3. The average admission time for each patient is 1.6. The average duration of each hospitalization is 14.9 days. For each admission, the static features include annotations on 13 laboratory tests, 17 diagnoses together with age and gender. There are time-series data of blood glucose measurements and 12 medications. The median of the number of daily blood glucose measurements is 4. The number of medication types per patient per day has a maximum of 12 and a mean of 3.1. In our experiment, the train/test set is split as 8:2. We perform 5-fold cross-validation under this ratio and report the mean of the results.

B. Experimental Setup

1) Baseline Methods: We compare our MERITS with three time-series medication recommendation models as baselines. As our data consists of irregular time-series dynamics within each observation t, the baselines are appropriately modified to fit our problem.

- **GRU†:** Gated Recurrent Unit (GRU) [13] as an encoder network whose input is the sequential dynamic embedding of patient's biomarkers, and medications concatenate by the static features.
- **RETAIN†:** RETAIN [14] is a two-level neural attention model that predicts medication combination and diseases by detecting influential past visits and clinical variables within those visits. We modify the inputs embedding with an RNN to process the dynamic features.
- **GAMENet†:** GAMENet [4] is a graph augmented model that recommends medication based on a memory neural network by leveraging EMR and Drug-Drug Interaction. We also modify the inputs part of the model using an RNN to process the dynamic features.

TABLE I
Performance comparison of different methods.

Method	DDI-R	Jaccard	PR-AUC	Pre	Rec	F1
GRU†	0.097	0.848	0.873	0.925	0.900	0.900
RETAIN†	0.089	0.878	0.899	0.944	0.921	0.922
GAMENet†	0.085	0.904	0.978	0.949	0.947	0.939
MERITS	**0.083**	**0.917**	**0.978**	**0.957**	**0.954**	**0.948**

2) Implementation Details: For all methods, the model input is the sequential observations (patient's static features, dynamic features, and medication history) from day 1 to day t, and output the medication prediction of day $t+1$. Each hospital day of a patient is considered as a sample.

For MERITS, we do grid search on the hyper-parameters and evaluate on the validation set. We search the tolerance DDI rate s from $\{0.03, 0.05, 0.10, 0.15\}$, the initial temperature $Temp$ from $\{0.3, 0.5, 0.7, 0.9\}$, the weight decay factor ϵ from $\{0.95, 0.90, 0.85, 0.8\}$, and the trade-off factor λ in the loss function from $\{0.9, 0.8, 0.7, 0.6\}$. For the sampler of Neural ODE, we select the irregular time sample combinations from the set $\{8, 12, 16, 20, 24\}$, which means the i-th hour in a day, e.g., 16 means 4 pm. The MERITS is optimized with Adam optimizer [15] with learning rate from the set $\{1e^{-4}, 3e^{-4}, 5e^{-4}, 7e^{-4}\}$. The batch size is 32. All models are trained/tested on a single Nvidia V100 GPU with 32GB memory. The source code is available at https://github.com/im73/MERITS.

3) Evaluation Metrics: We evaluate the performance of the comparison methods using Average F1 score (F1), Precision-Recall AUC (PR-AUC) and Jaccard Similarity Score (Jaccard).

C. Overall Performance

Table I lists the overall performance of MERITS and baseline methods predicting medication on the diabetic inpatient medication dataset. The experiment results show that MERITS consistently achieves better performance than the baseline methods on most metrics. Specifically, there are several findings: (1) MERITS outperforms all baseline approaches. This demonstrates the success of the framework and the effectiveness of the irregular time-series encoding and the attention mechanisms. (2) The DDI Rate of MERITS and GAMENet† is lower than other baselines, which indicates that MERITS learns the knowledge of DDI representation from the DDI graph that improves the medication safety. (3) The graph augmented methods MERITS and GAMENet† perform better

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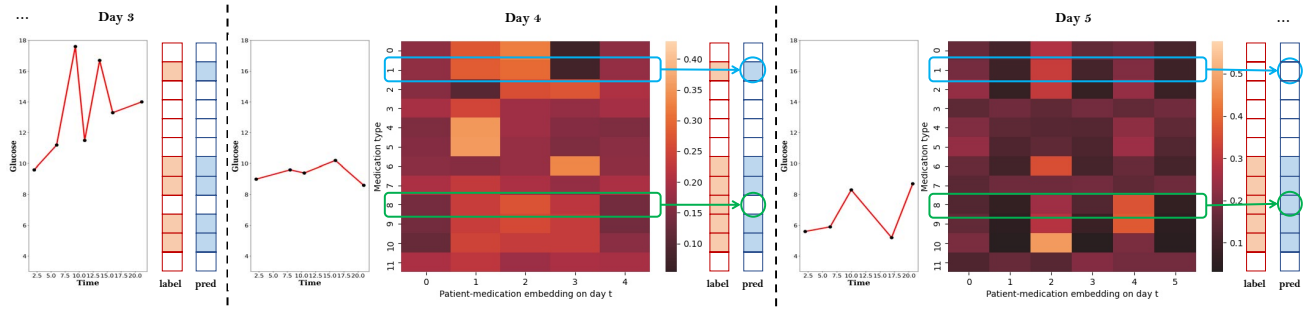


Fig. 3. Visualization of the multi-head self-attention in the decoder

than the methods without graph information, which implies it is useful to guide the model with knowledge and our model makes good use of such information. (4) MERITS achieve better performance than GAMENet†, which indicates that only considering the DDI graph may restrict some medication rules by external knowledge. The other two graphs in MERITS designed for capturing the history medication pattern can alleviate this restriction by learning about historical medication.

D. Case Study: Visualization

In order to visually observe the effect of MERITS, we select one typical patient from the test set whose medication changes in several adjacent days. In this way, we show how our model MERITS can predict correct medication recommendations based on the changes in the patient's EMR. We visualize the multi-head self-attention in the decoder for two adjacent days in Figure 3. The attention combines the patient-medication embedding $\boldsymbol{E}_{en}$ and the graph embedding of historical medications $\boldsymbol{E}_G$. This figure shows that the patient's glucose goes down from Day 3 to Day 4, which means the medication strategy in Day 3 is efficient. However, the medication label changed on Day 4 due to some reason, and our MERITS missed the Drug 8 (the green circle in Figure 3). With the new medication strategy, the patient's glucose remains stable on Day 5, so MERITS also orders Drug 8. Meanwhile, with the new calculated attention, Drug 1 attends less on the patient-medication embedding on Day 5, so MERITS does not predict Drug 1 on Day 5.

V. Conclusion

This paper aims to provide solutions to medication recommendations for chronic disease patients with complex historical electronic medical records (EMR). We propose MERITS, an attention-based encoder-decoder framework, to address the irregular time-series dependencies problem and continuous correlation problem. The experiments on our collect medical dataset demonstrate the effectiveness of MERITS. The case study and visualization show the interpretability of the model.

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References

[1] Z. C. Lipton, D. Kale, and R. Wetzel, "Directly modeling missing data in sequences with rnns: Improved classification of clinical time series," in *Machine learning for healthcare conference*. PMLR, 2016, pp. 253–270.

[2] T. Q. Chen, Y. Rubanova, J. Bettencourt, and D. Duvenaud, "Neural ordinary differential equations," in *Annual Conference on Neural Information Processing Systems 2018*, 2018, pp. 6572–6583.

[3] Y. He, C. Wang, N. Li, and Z. Zeng, "Attention and memory-augmented networks for dual-view sequential learning," in *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2020, pp. 125–134.

[4] J. Shang, C. Xiao, T. Ma, H. Li, and J. Sun, "Gamenet: Graph augmented memory networks for recommending medication combination," in *proceedings of the AAAI Conference on Artificial Intelligence*, vol. 33, no. 01, 2019, pp. 1126–1133.

[5] J. Shang, T. Ma, C. Xiao, and J. Sun, "Pre-training of graph augmented transformers for medication recommendation," in *Proceedings of the Twenty-Eighth International Joint Conference on Artificial Intelligence*, S. Kraus, Ed. ijcai.org, 2019, pp. 5953–5959.

[6] L. Wang, W. Yu, X. He, W. Cheng, M. R. Ren, W. Wang, B. Zong, H. Chen, and H. Zha, "Adversarial cooperative imitation learning for dynamic treatment regimes," in *Proceedings of The Web Conference 2020*, 2020, pp. 1785–1795.

[7] L. Wang, W. Zhang, X. He, and H. Zha, "Supervised reinforcement learning with recurrent neural network for dynamic treatment recommendation," in *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2018, pp. 2447–2456.

[8] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," in *Annual Conference on Neural Information Processing Systems 2017*, 2017, pp. 5998–6008.

[9] D. E. Rumelhart, G. E. Hinton, and R. J. Williams, "Learning internal representations by error propagation," California Univ San Diego La Jolla Inst for Cognitive Science, Tech. Rep., 1985.

[10] P. Velickovic, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio, "Graph attention networks," in *6th International Conference on Learning Representations*. OpenReview.net, 2018.

[11] B. Xu, N. Wang, T. Chen, and M. Li, "Empirical evaluation of rectified activations in convolutional network," *arXiv preprint arXiv:1505.00853*, 2015.

[12] S. Kirkpatrick, C. D. Gelatt, and M. P. Vecchi, "Optimization by simulated annealing," *Science*, vol. 220, no. 4598, pp. 671–680, 1983.

[13] J. Chung, C. Gulcehre, K. Cho, and Y. Bengio, "Empirical evaluation of gated recurrent neural networks on sequence modeling," *arXiv preprint arXiv:1412.3555*, 2014.

[14] E. Choi, M. T. Bahadori, J. Sun, J. Kulas, A. Schuetz, and W. F. Stewart, "RETAIN: an interpretable predictive model for healthcare using reverse time attention mechanism," in *Annual Conference on Neural Information Processing Systems 2016*, 2016, pp. 3504–3512.

[15] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *3rd International Conference on Learning Representations*, Y. Bengio and Y. LeCun, Eds., 2015.

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